

1 Helmholtz's theorem; scalar potential; Earnshaw's theorem

- The four Maxwell equations are tied together by time derivatives. Without time dependence (and current, so we can neglect the two equations describing $\mathbf{B}$), these read

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}, \quad \nabla \times \mathbf{E} = 0.$$

This puts us in the *electrostatic* regime, where the only quantity of interest is the electric field $\mathbf{E}$. Broadly speaking, we encounter two classes of problem:

- (1) *Summation problems*, where the charge density $\rho(\mathbf{r})$ is specified everywhere, and you do an integral to determine the field.
- (2) *Boundary-value problems*, where in general $\rho(\mathbf{r})$ needs to be determined, and you solve the equations

$$\nabla^2 \varphi = 0, \quad \nabla^2 \varphi = -\rho/\epsilon_0,$$

subject to the appropriate boundary conditions.

In this regime, the Lorentz force law can be written as $\mathbf{F} = q\mathbf{E}$. A nice generalization that goes beyond point charges is

$$\mathbf{F} = \int d^3r \rho^*(\mathbf{r})\mathbf{E}(\mathbf{r}), \quad (1.1)$$

where ρ^* represents the charge distribution being acted upon by an external field $\mathbf{E}$. The instructor prefers this form of writing.

- Now before moving on to the physical question of determining $\mathbf{E}$, let us review some math. *Helmholtz's theorem* states that any vector field can be decomposed into the sum of a divergence-free and a curl-free field:

$$\mathbf{C} = \mathbf{C}_\perp + \mathbf{C}_\parallel,$$

where $\nabla \cdot \mathbf{C}_\perp = 0$ and $\nabla \times \mathbf{C}_\parallel = 0$. The same theorem implies that there exist vector and scalar fields $\mathbf{F}$ and Ω respectively, such that $\mathbf{C}_\perp = \nabla \times \mathbf{F}$ and $\mathbf{C}_\parallel = -\nabla\Omega$, which are given by

$$\Omega(\mathbf{r}) = \frac{1}{4\pi} \int d^3r' \frac{\nabla' \cdot \mathbf{C}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}, \quad \mathbf{F}(\mathbf{r}) = \frac{1}{4\pi} \int d^3r' \frac{\nabla' \times \mathbf{C}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}.$$

Thus, for an arbitrary field $\mathbf{E}$ we can write

$$\begin{aligned} \mathbf{E} &= \mathbf{E}_\perp + \mathbf{E}_\parallel \\ &= -\nabla \left[\frac{1}{4\pi} \int d^3r' \frac{\nabla' \cdot \mathbf{E}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \right] + \nabla \times \left[\frac{1}{4\pi} \int d^3r' \frac{\nabla' \times \mathbf{E}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \right]. \end{aligned}$$

- Combining this purely mathematical result with Maxwell's equations, we obtain¹

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \int d^3r' \rho(\mathbf{r}') \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|^3}. \quad (1.2)$$

Given $\rho(\mathbf{r})$, you can, in principle, compute the electric field using this formula. Then by substituting this back into our expression for force (1.1), we obtain for the force on a charge distribution ρ^* the result

$$\mathbf{F} = \frac{1}{4\pi\epsilon_0} \int d^3r \int d^3r' \rho^*(\mathbf{r})\rho(\mathbf{r}') \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|^3}. \quad (1.3)$$

¹Note that the derivative operators outside the square brackets are acting on the *unprimed* variables, so we are free to move these inside the integrals.

We see by inspection that this obey's Newton's third law; in particular, a charge distribution cannot exert a net force on itself. Returning to (1.2), we can say that we've "solved" all summation problems, but this "solution" is a formal integral, which is to say that evaluating it is hopelessly impractical in many situations.

- There is an alternative approach to obtaining (1.2) which doesn't invoke Helmholtz's theorem directly. Consider an arrangement of n point charges q_i , with $i = 1, \dots, n$. We start with Coulomb's law and assume that the electric field obeys the principle of *superposition*, i.e., the field from a given q_i is not changed by the presence of other fields. Then since $\mathbf{F} = q\mathbf{E}$, we get

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_i q_i \frac{\mathbf{r} - \mathbf{r}_i}{|\mathbf{r} - \mathbf{r}_i|^3},$$

and by making the identification $\sum_i q_i \rightarrow \int d^3\mathbf{r}' \rho(\mathbf{r}')$ we obtain (1.2), as earlier.

- It is often more convenient to work with the *electrostatic scalar potential*. By Helmholtz's theorem, since $\nabla \times \mathbf{E} = 0$, we can write $\mathbf{E} = -\nabla\varphi$, where φ is a scalar field given by

$$\varphi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int d^3r' \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}. \quad (1.4)$$

Indeed, in practice this integral is easier to evaluate, since there are no vectors involved. There is some arbitrariness in the definition of the scalar potential, since the physical laws are unchanged for $\varphi(\mathbf{r}) \rightarrow \varphi(\mathbf{r}) + c$, where c is a constant. Other choices can be convenient, as we'll see later when we discuss gauge transformations.

- The Coulomb force is conservative: consider the work done by the electrostatic field on a charge from $\mathbf{r}_1$ to $\mathbf{r}_2$. We have

$$W = q \int_{\mathbf{r}_1}^{\mathbf{r}_2} \mathbf{E} \cdot d\boldsymbol{\ell} = -q \int_{\mathbf{r}_1}^{\mathbf{r}_2} (\nabla\varphi) \cdot d\boldsymbol{\ell} = -q[\varphi(\mathbf{r}_2) - \varphi(\mathbf{r}_1)],$$

which is manifestly path-independent. Equivalently, the work done over any closed loop is zero. This follows because the electrostatic field is curl-free:

$$W = q \oint \mathbf{E} \cdot d\boldsymbol{\ell} = q \int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{S} = 0.$$

- In electrostatics, we have *Earnshaw's theorem*, which states that for any finite, charge-free region R , the maximum or minimum values of the scalar potential $\varphi(\mathbf{r})$ occur at the boundary ∂R . Equivalently, it is impossible to construct an electrostatic field that will hold a charged particle in stable equilibrium.

Proof. Suppose that $\varphi(\mathbf{r})$ has a minimum inside R ; that is, suppose $\nabla\varphi = 0$ at a point P in R . Now consider an arbitrary small surface S enclosing P , and let $\hat{\mathbf{n}}$ denote the outward-pointing normal to S . Since P is a minimum, we have $\hat{\mathbf{n}} \cdot \nabla\varphi > 0$ everywhere on S . Thus

$$\int_S dS \hat{\mathbf{n}} \cdot \nabla\varphi > 0 \implies \int_V d^3r \nabla \cdot (\nabla\varphi) > 0 \implies \int_V d^3r \frac{\rho}{\epsilon_0} < 0,$$

contradicting our assumption that $\rho(\mathbf{r}) = 0$ everywhere in R . A similar argument applies for the case where P is a maximum. $\square$

Evidently, one needs more than a configuration of static charges to produce stable matter. A more complete picture is offered by quantum mechanics!

2 Electrostatic energy

- The *total electrostatic energy* U_E is the work it takes to assemble a configuration of charges by bringing them in from infinity. One can write this as a sum of pairwise potentials:

$$U_E = \frac{1}{8\pi\epsilon_0} \sum_{i=1}^N \sum_{j \neq i}^N \frac{q_i q_j}{|r_i - r_j|} = \frac{1}{2} \sum_{i=1}^N q_i \varphi(\mathbf{r}_i). \quad (2.1)$$

The factor of $1/2$ accounts for double-counting, while the requirement $j \neq i$ excludes contributions from self-energy. The continuous analog of this expression is

$$U_E = \frac{1}{2} \int d^3r \rho(\mathbf{r}) \varphi(\mathbf{r}). \quad (2.2)$$

One needs to take care when computing U_E for a system containing point charges; their infinite self-energy needs to be excluded by hand. This problem persists even if we try a limiting procedure. Consider a solid sphere of uniformly distributed charge Q and radius R . It can be shown that

$$U_E = \frac{3}{5} \frac{Q^2}{4\pi\epsilon_0 R},$$

which also diverges as $R \rightarrow 0$. This is, in fact, a fundamental limitation of the point charge concept in classical electrodynamics.

- Using Gauss' law and assuming that $\mathbf{E} \rightarrow 0$ at infinity, we can rewrite (2.2) to get

$$\begin{aligned} U_E &= \frac{1}{2} \int d^3r \rho(\mathbf{r}) \varphi(\mathbf{r}) = \frac{\epsilon_0}{2} \int d^3r (\nabla \cdot \mathbf{E}) \varphi(\mathbf{r}) \\ &= \frac{\epsilon_0}{2} \int d^3r \nabla \cdot (\varphi \mathbf{E}) - \frac{\epsilon_0}{2} \int d^3r \mathbf{E} \cdot (\nabla \varphi) \\ &= \frac{\epsilon_0}{2} \oint dS \hat{\mathbf{n}} \cdot (\varphi \mathbf{E}) + \frac{\epsilon_0}{2} \int d^3r |\mathbf{E}|^2 \\ &= \frac{\epsilon_0}{2} \int d^3r |\mathbf{E}|^2. \end{aligned} \quad (2.3)$$

In this form it is evident that $U_E \geq 0$. Here we also define the *electrostatic energy density*

$$u_E = \frac{\epsilon_0}{2} |\mathbf{E}|^2, \quad (2.4)$$

which is a scalar field. Once upon a time, it was unclear whether the Coulomb force or the electric field was to be thought of as more fundamental. This result shows (or at least suggests) that one should think of $\mathbf{E}$ as its own entity, carrying its own energy and so forth.

- You may come across a “contradiction” if you consider two point charges $q_1 < 0$ and $q_2 > 0$, for which (2.1) gives

$$U_E = \frac{q_1 q_2}{4\pi\epsilon_0 |r_1 - r_2|} < 0.$$

But we just asserted that $U_E \geq 0$, so what happened? The answer lies in the fact that, somewhat confusingly, (2.1) and (2.3) are slightly different quantities. The former definition excludes self-energy terms via the prescription $j \neq i$ in the double sum, but the latter does not. While these terms are formally divergent, they suffice (we are told) to make $U_E \geq 0$.²

²Aside from this minor point, the two expressions are equivalent; see Problem 1.33 in Purcell and Morin for further discussion.

3 The electric stress tensor; electric multipoles

- Suppose we're given a charge density distribution $\rho(\mathbf{r})$ occupying some region Ω , and suppose that $\mathbf{E}(\mathbf{r})$ is the electric field due to this distribution. Then the charge density $\rho(\mathbf{r} \in V)$ experiences a force due to $\rho(\mathbf{r} \in \Omega - V)$ which is given by

$$\mathbf{F} = \int_V d^3r \rho(\mathbf{r}) \mathbf{E}(\mathbf{r}).$$

Using the first of Maxwell's equations, this becomes

$$\mathbf{F} = \epsilon_0 \int_V d^3r (\nabla \cdot \mathbf{E}) \mathbf{E},$$

which has components

$$\begin{aligned} F_j &= \epsilon_0 \int_V d^3r E_j \partial_i E_i \\ &= \epsilon_0 \int_V d^3r [\partial_i (E_j E_i) - E_i \partial_i E_j]. \end{aligned}$$

The other Maxwell equation $\nabla \times \mathbf{E} = 0$ implies that $\partial_i E_j - \partial_j E_i = 0$, so we can exchange indices in the second term to get

$$\begin{aligned} F_j &= \epsilon_0 \int_V d^3r [\partial_i (E_j E_i) - E_i \partial_j E_i] \\ &= \epsilon_0 \int_V d^3r \partial_i \left(E_i E_j - \frac{1}{2} \delta_{ij} |\mathbf{E}|^2 \right). \end{aligned}$$

This motivates the definition of the *electric stress tensor*,

$$T_{ij} = \epsilon_0 \left(E_i E_j - \frac{1}{2} \delta_{ij} |\mathbf{E}|^2 \right). \quad (3.1)$$

- If we apply the divergence theorem to the previous expression, then we get

$$F_j = \int_S dS \hat{n}_i T_{ij},$$

with $\hat{n}_i$ representing the i th component of the unit vector normal to S . This shows that T_{ij} has dimensions of *stress*, or force per unit area, explaining the terminology. It also allows us to define a vector force density $\mathbf{f}$, where $f_j \equiv \hat{n}_i T_{ij}$, which describes the net force transmitted through a surface element $\hat{\mathbf{n}} dS$.

- We define electric multipoles. In a sense, we are now interested in “long-distance” physics; we do not concern ourselves with what happens very close to a given charge distribution. To illustrate the idea, consider some arbitrary, lumpy charge distribution with nonzero net charge Q . At nearby points the field might be very complicated, but as one moves further out, the field begins to approach that of a point charge. Equivalently, at large distances

$$\varphi(\mathbf{r}) = \frac{Q}{4\pi\epsilon_0 r} + \text{corrections}. \quad (3.2)$$

To be more precise about this, we need to define what we mean by “far away.” First, we limit ourselves to physical scenarios by assuming that the charge distribution is localized, i.e., there is some sphere of finite radius R that can enclose it. This gives us a length scale to work with.

- The idea is fairly simple. We start from the definition (1.4) for electrostatic potential, then expand the term $1/|\mathbf{r} - \mathbf{r}'|$ in a Taylor series around r'/r , which is valid because $r \gg R > r'$ by assumption. This gives us the *electrostatic multipole expansion*

$$\varphi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \left\{ \left[\int d^3r' \rho(\mathbf{r}') \right] \frac{1}{r} - \left[\int d^3r' \rho(\mathbf{r}') r'_i \right] \nabla_i \frac{1}{r} + \left[\frac{1}{2} \int d^3r' \rho(\mathbf{r}') r'_i r'_j \right] \nabla_i \nabla_j \frac{1}{r} + \dots \right\}$$

and further terms can be obtained systematically by computing higher-order terms in the Taylor expansion. The expressions in brackets are multipole moments: the *electric monopole moment* is just the total charge

$$Q = \int d^3r \rho(\mathbf{r}),$$

the *electric dipole moment* is the vector

$$\mathbf{p} = \int d^3r \rho(\mathbf{r}) \mathbf{r},$$

and the *electric quadrupole moment* is a rank two tensor with components

$$Q_{ij} = \frac{1}{2} \int d^3r \rho(\mathbf{r}) r_i r_j.$$

And so forth. A quick order of magnitude estimate shows that each term in the multipole expansion is smaller than the one that precedes it by a factor $R/r \ll 1$. Thus, the behavior at long distances is dominated by the contribution from the first nonzero multipole.

- Using the multipole definitions above and evaluating the derivatives, we obtain

$$\varphi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \left\{ \frac{Q}{r} + \frac{\mathbf{p} \cdot \mathbf{r}}{r^3} + Q_{ij} \frac{3r_i r_j - r^2 \delta_{ij}}{r^5} + \dots \right\}, \quad (3.3)$$

which is also referred to as the *primitive Cartesian electric multipole expansion*. This constitutes a formal elaboration of the idea in Eq. (3.2). In a later lecture, we'll derive an analogous expansion which is more appropriate for spherical coordinates.

4 Dipoles and quadrupoles

- We derive an expression for the potential energy of a point dipole located at a point $\mathbf{r}$ in an external field $\mathbf{E}$. Assuming $\mathbf{p}$ is constant, we get

$$\begin{aligned} V_E(\mathbf{r}) &= \int d^3r' \varphi(\mathbf{r}') \rho_D(\mathbf{r}') \\ &= \int d^3r' \varphi(\mathbf{r}') [-\mathbf{p} \cdot \nabla' \delta(\mathbf{r}' - \mathbf{r})] \\ &= -\mathbf{p} \cdot \int d^3r' \nabla' [\varphi(\mathbf{r}') \delta(\mathbf{r} - \mathbf{r}')] + \mathbf{p} \cdot \int d^3r' \delta(\mathbf{r}' - \mathbf{r}) \nabla' [\varphi(\mathbf{r}')] \\ &= -\mathbf{p} \cdot \mathbf{E}(\mathbf{r}), \end{aligned}$$

where we've integrated by parts to show that the first integral on the third line is zero. In particular, for a point dipole with constant $\mathbf{p}$, the net force exerted upon it by an external field is $\mathbf{F} = \nabla(\mathbf{p} \cdot \mathbf{E})$, so that one needs an electric field gradient to generate a force.

- We give the electrostatic energy for a configuration of point dipoles. Consider first the work done by dipole $\mathbf{p}_1$ on dipole $\mathbf{p}_2$ when it is moved to the position $\mathbf{r}_2$. From our previous result,

$$\begin{aligned} W_{12} &= -\mathbf{p}_2 \cdot \mathbf{E}_1(\mathbf{r}_2) \\ &= \frac{1}{4\pi\epsilon_0} \left[\frac{\mathbf{p}_1 \cdot \mathbf{p}_2}{|\mathbf{r}_2 - \mathbf{r}_1|^3} - \frac{3[\mathbf{p}_1 \cdot (\mathbf{r}_2 - \mathbf{r}_1)][\mathbf{p}_2 \cdot (\mathbf{r}_2 - \mathbf{r}_1)]}{|\mathbf{r}_2 - \mathbf{r}_1|^5} \right]. \end{aligned}$$

By superposition, the total electrostatic energy for a configuration of N point dipoles is

$$U_E = \frac{1}{4\pi\epsilon_0} \cdot \frac{1}{2} \sum_{i=1}^N \sum_{j \neq i}^N \left[\frac{\mathbf{p}_i \cdot \mathbf{p}_j}{|\mathbf{r}_i - \mathbf{r}_j|^3} - \frac{3[\mathbf{p}_i \cdot (\mathbf{r}_i - \mathbf{r}_j)][\mathbf{p}_j \cdot (\mathbf{r}_i - \mathbf{r}_j)]}{|\mathbf{r}_i - \mathbf{r}_j|^5} \right].$$

This discussion is taking more time than the instructor expected, but the point here is that there's more to electric multipoles than what we see in Griffiths. In particular, a quadrupole is more than just an arrangement of equal and opposite point charges at the corners of a cube (although this does produce a charge distribution whose first nonvanishing moment is at quadrupole order). All extended bodies, in general, will see some quadrupole contribution to the asymptotic field.

- In nature, objects with zero total charge and zero dipole moment are extremely common. For instance, the latter happens whenever you have spherical symmetry, because then the “centers” of positive and negative charge coincide at the origin. This motivates the study of the *electric quadrupole*, which contributes to the electrostatic potential like

$$\frac{1}{4\pi\epsilon_0} Q_{ij} \frac{3r_i r_j - \delta_{ij} r^2}{r^5}.$$

We construct a point quadrupole by bringing together two point dipoles with opposite moments $\pm\mathbf{p}/s$, separated by a vector $s\mathbf{c}$ (picture necessary), and then performing a Taylor expansion:

$$\nabla \frac{1}{|\mathbf{r} - \mathbf{r}_0 - s\mathbf{c}|} \approx \nabla \left[\frac{1}{|\mathbf{r} - \mathbf{r}_0|} - s(\mathbf{c} \cdot \nabla) \frac{1}{|\mathbf{r} - \mathbf{r}_0|^2} \right].$$

In the limit $s \rightarrow 0$, we obtain the expression

$$\begin{aligned} \varphi(\mathbf{r}) &= \frac{1}{4\pi\epsilon_0} (\mathbf{p} \cdot \nabla)(\mathbf{c} \cdot \nabla) \frac{1}{|\mathbf{r} - \mathbf{r}_0|} \\ &= \frac{1}{4\pi\epsilon_0} p_i c_j \nabla_i \nabla_j \frac{1}{|\mathbf{r} - \mathbf{r}_0|}, \end{aligned}$$

which suggests that we write $Q_{ij} = p_i c_j$. But the quadrupole tensor is symmetric by definition, so we must symmetrize, which leaves us with

$$Q_{ij} = \frac{1}{2}(c_i p_j + c_j p_i).$$

The way to think about Q_{ij} is as a matrix, since it's a rank two tensor. The symmetry implies that there are only $3(3+1)/2 = 6$ independent components, and also that Q_{ij} is diagonalizable.

- While this stuff is important for seeing *where* you get contributions to the potential, it won't be very useful if you cannot do the integrals. Thus, sometimes it is easier to work with the spherical expansion, which we'll discuss next lecture.
- Now we introduce a traceless quadrupole tensor.³ This development reduces the number of components you need by one, since $Q_{xx} + Q_{yy} + Q_{zz} = 0$ constrains one of the diagonals. Recall that the quadrupole contribution to the potential is proportional to

$$\left\{ \frac{1}{2} \int d^3 r' \rho(\mathbf{r}') r'_i r'_j \right\} \frac{3r_i r_j - \delta_{ij} r^2}{r^5}.$$

The goal here is to get something like

$$\left\{ \dots \right\}_{ij} \frac{r_i r_j}{r^5}.$$

Now since $r'_i r'_j r^2 \delta_{ij} = r^2 r'^2 = r'^2 r_i r_j \delta_{ij}$, we can rewrite the previous expression as

$$\left\{ \frac{1}{2} \int d^3 r' \rho(\mathbf{r}') (3r'_i r'_j - \delta_{ij} r'^2) \right\} \frac{r_i r_j}{r^5}.$$

While this may look like a superficial rewriting, it allows us to define a new two-index object, which we call the *traceless quadrupole tensor*:

$$\Theta_{ij} = \frac{1}{2} \int d^3 r' \rho(\mathbf{r}') (3r'_i r'_j - \delta_{ij} r'^2).$$

This is traceless because $\delta_{ii} = 3$, and therefore it has $6 - 1 = 5$ independent components.

- Consider an electric field $\mathbf{E}$ which is slowly varying in space around some point $\mathbf{r}$, and then examine nearby points $\mathbf{r}'$:

$$\mathbf{E}(\mathbf{r}') = \mathbf{E}(\mathbf{r}) + [(\mathbf{r}' - \mathbf{r}) \cdot \nabla] \mathbf{E} + \frac{1}{2} [(\mathbf{r}' - \mathbf{r}) \cdot \nabla]^2 \mathbf{E} + \dots$$

The goal is to compute the net force $\mathbf{F}$ on some charge distribution $\rho(\mathbf{r}')$ localized around $\mathbf{r}$,

$$\mathbf{F} = \int d^3 r' \rho(\mathbf{r}') \mathbf{E}(\mathbf{r}').$$

This is easiest to do using index notation. The first term brings a contribution from Q alone, the second term involves both Q and the dipole moment $\mathbf{p}$, and the third term brings in terms like $r'_i r'_j \nabla_i \nabla_j E_k$, which results in a contribution from the quadrupole Q_{ij} . For a point quadrupole, the result is nice and simple:

$$\mathbf{F} = Q_{ij} \nabla_i \nabla_j \mathbf{E}.$$

Meanwhile, the torque $\mathbf{N}$ is given by (justify)

$$\mathbf{N} = 2(\mathbf{Q} \cdot \nabla) \times \mathbf{E} + \mathbf{r} \times \mathbf{F}.$$

³Jackson simply refers to this as *the quadrupole tensor*, so bear that in mind when moving between references.

5 Spherical multipoles; trace-free electric multipole tensors

- The multipole moments we've discussed so far look like

$$\int d^3r \rho(\mathbf{r}) r_i r_j \cdots r_k$$

which are fine, provided you can evaluate these integrals. But what if you have spherical symmetry, so that $\rho(\mathbf{r}) = \rho(r, \theta, \phi)$ is a more natural way to describe your system? Let us try another expansion of $1/|\mathbf{r} - \mathbf{r}'|$, now with a view toward spherical coordinates. First note that

$$\frac{1}{|\mathbf{r} - \mathbf{r}'|} = \frac{1}{\sqrt{r^2 + r'^2 + 2\mathbf{r} \cdot \mathbf{r}'}} = \frac{1}{r} \frac{1}{\sqrt{1 + (r'/r)^2 - 2(r'/r)\underbrace{(\hat{\mathbf{r}} \cdot \hat{\mathbf{r}}')}_{\cos \gamma}}},$$

where γ is the angle formed by the unit vectors $\hat{\mathbf{r}}$ and $\hat{\mathbf{r}}'$. Now, this is a generating function:

$$\frac{1}{\sqrt{1 - 2xt + t^2}} = \sum_{\ell=0}^{\infty} P_{\ell}(x) t^{\ell},$$

where $P_{\ell}(x)$ is the Legendre polynomial of order ℓ . This is valid provided $|x| \leq 1$ and $0 < t < 1$, both of which are true for our expression. Thus, we have the alternative expansion

$$\frac{1}{|\mathbf{r} - \mathbf{r}'|} = \frac{1}{r} \sum_{\ell=0}^{\infty} P_{\ell}(\hat{\mathbf{r}} \cdot \hat{\mathbf{r}}') \left(\frac{r'}{r}\right)^{\ell} \quad (r' < r).$$

The argument $\hat{\mathbf{r}} \cdot \hat{\mathbf{r}}'$ is inconvenient for calculations, but we can simplify things using the *spherical harmonic addition theorem*,

$$P_{\ell}(\hat{\mathbf{r}} \cdot \hat{\mathbf{r}}') = \frac{4\pi}{2\ell + 1} \sum_{m=-\ell}^{\ell} Y_{\ell m}^*(\theta', \phi') Y_{\ell m}(\theta, \phi).$$

This yields

$$\frac{1}{|\mathbf{r} - \mathbf{r}'|} = \frac{1}{r} \sum_{\ell=0}^{\infty} \frac{4\pi}{2\ell + 1} \left(\frac{r'}{r}\right)^{\ell} \sum_{m=-\ell}^{\ell} Y_{\ell m}^*(\theta', \phi') Y_{\ell m}(\theta, \phi) \quad (r' < r).$$

If we now introduce the notation $r_{<}$ and $r_{>}$ for the smaller and larger term, respectively, of r and r' , then we get the compact expression

$$\frac{1}{|\mathbf{r} - \mathbf{r}'|} = \frac{1}{r_{>}} \sum_{\ell=0}^{\infty} \frac{4\pi}{2\ell + 1} \left(\frac{r_{<}}{r_{>}}\right)^{\ell} \sum_{m=-\ell}^{\ell} Y_{\ell m}^*(\theta_{<}, \phi_{<}) Y_{\ell m}(\theta_{>}, \phi_{>}).$$

The reason for this rewriting is that it allows us to examine two regimes. If the source points are such that $r' < r$, then we say we have an *exterior spherical expansion*, and if $r' > r$, then we have an *interior spherical expansion*. The former is more common (it corresponds to all the charge being localized to some sphere of radius R), so we'll focus on that.

- Substituting this back into the Coulomb integral (1.4), we obtain

$$\varphi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \left\{ \frac{4\pi}{2\ell + 1} \int d^3r' \rho(\mathbf{r}') (r')^{\ell} Y_{\ell m}^*(\theta', \phi') \right\} \frac{Y_{\ell m}(\theta, \phi)}{r^{\ell+1}}.$$

If we now define the *exterior spherical multipole moments* by

$$A_{\ell m} \equiv \frac{4\pi}{2\ell+1} \int d^3r' \rho(\mathbf{r}') (r')^\ell Y_{\ell m}^*(\theta', \phi'),$$

then we have an expansion of the electric potential in spherical coordinates:

$$\varphi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} A_{\ell m} \frac{Y_{\ell m}(\theta, \phi)}{r^{\ell+1}}. \quad (5.1)$$

This is analogous to the expansion (3.3) in primitive Cartesian multipoles.

- A quick back of the envelope estimate tells us that $A_{\ell m} \sim QR^\ell$, so the ℓ -th order term in the spherical expansion goes like

$$QR^\ell \frac{1}{r^{\ell+1}} \sim \frac{Q}{r} \left(\frac{R}{r} \right)^\ell,$$

showing that the contributions decrease with increasing order, as in the Cartesian expansion. To show that the two expansions are indeed equivalent, we write out (5.1) explicitly:

$$\varphi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \left\{ \frac{A_{00}Y_{00}(\Omega)}{r} + \frac{A_{1,-1}Y_{1,-1}(\Omega) + A_{1,0}Y_{1,0}(\Omega) + A_{1,1}Y_{1,1}(\Omega)}{r^2} + \dots \right\}$$

By transforming from spherical to Cartesian coordinates, one can show that $A_{00}Y_{00}(\Omega) = Q$ and $A_{1,-1}Y_{1,-1}(\Omega) + \dots + A_{1,1}Y_{1,1}(\Omega) = \mathbf{p} \cdot \hat{\mathbf{r}}$, which shows that the same information is encoded at each order in the spherical and Cartesian expansions.

- Notice that one only needs to calculate $2\ell + 1$ moments at each order, rather than the $\frac{1}{2}(\ell + 2)(\ell + 1)$ required by the primitive Cartesian expansion.⁴ Thus, one might be led to believe that the spherical expansion is superior, as the difference in number of moments at higher orders becomes significant. However, this can be addressed by the trace-free formulation of the Cartesian multipole tensors – which we actually got a hint of earlier, when we introduced the tensor Θ_{ij} . If we define the tensors

$$T_{ij\dots m}^{(\ell)} = \frac{(-1)^\ell}{\ell!} \int d^3y \rho(\mathbf{y}) y^{2\ell+1} \frac{\partial}{\partial y_i} \frac{\partial}{\partial y_j} \dots \frac{\partial}{\partial y_m} \frac{1}{|\mathbf{y}|},$$

which are trace-free because $\delta_{ij}\partial_i\partial_j(1/r) = -4\pi\delta_{(3)}(\mathbf{y})$, then we can write

$$\varphi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_{\ell=0}^{\infty} T_{ij\dots m}^{(\ell)} \frac{r_i r_j \dots r_m}{r^{2\ell+1}}.$$

Then since $T_{ij\dots m}^{(\ell)}$ are symmetric and trace-free, they have

$$\frac{1}{2}(\ell + 2)(\ell + 1) - \binom{\ell}{2} = 2\ell + 1$$

independent components, the same as with the spherical expansion. The choice of spherical or Cartesian coordinates is a matter of convenience; as long as you work with the trace-free Cartesian tensors, the number of integrals necessary will be the same in either representation.

⁴This number follows from the fact that the ℓ -th order primitive Cartesian multipole tensor is symmetric in all ℓ indices; for a calculation, see Section 4.7 of Zangwill.